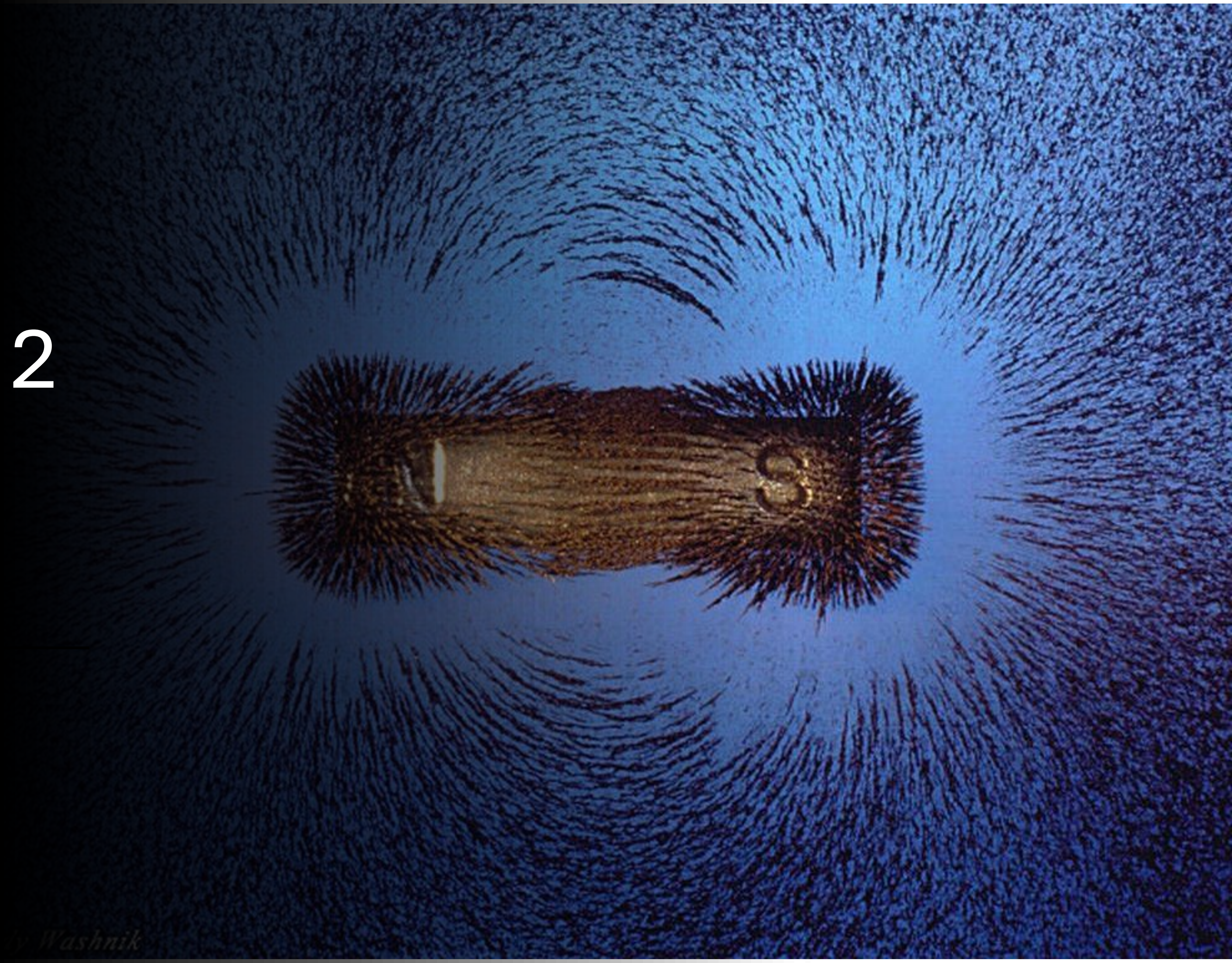
Mathematics 2 (5EZB0) Lecture 5



Koondi Mitra

Mathematics & Computer Science Department

Partial derivatives

Let the function f be continuous at a point

$$\boldsymbol{x} = (x_1, x_2, .., x_d) \in \mathcal{D}.$$

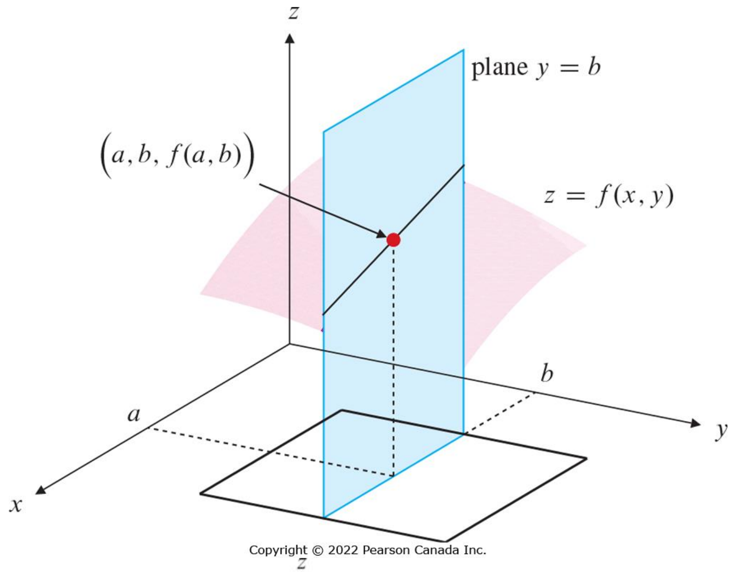
Its partial derivatives are then defined as

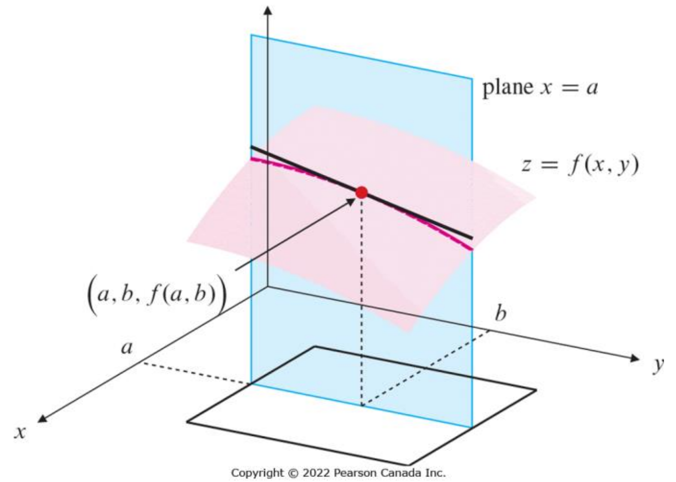
$$\frac{\partial f}{\partial x_1}(\boldsymbol{x}) = \lim_{\Delta x_1 \to 0} \frac{f(x_1 + \Delta x_1, x_2, .., x_d) - f(x_1, x_2, .., x_d)}{\Delta x_1}$$

$$\frac{\partial f}{\partial x_2}(\boldsymbol{x}) = \lim_{\Delta x_2 \to 0} \frac{f(x_1, x_2 + \Delta x_2, .., x_d) - f(x_1, x_2, .., x_d)}{\Delta x_2}$$

$$\vdots$$

If these limits exist.





Second order derivatives

Let the function f have continuous partial derivatives at a point $\boldsymbol{x} = (x_1, x_2, .., x_d) \in \mathcal{D}$.
Then its higher order derivatives are defined as

$$\frac{\partial^2 f}{\partial x_1^2}(\boldsymbol{x}) = \frac{\partial}{\partial x_1}\left(\frac{\partial f}{\partial x_1}\right)(\boldsymbol{x}) = \lim_{\Delta x_1 \to 0} \frac{\partial_{x_1} f(x_1 + \Delta x_1, x_2, .., x_d) - \partial_{x_1} f(x_1, x_2, .., x_d)}{\Delta x_1}$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_1}(\boldsymbol{x}) = \frac{\partial}{\partial x_2}\left(\frac{\partial f}{\partial x_1}\right)(\boldsymbol{x}) = \lim_{\Delta x_2 \to 0} \frac{\partial_{x_1} f(x_1, x_2 + \Delta x_2, .., x_d) - \partial_{x_1} f(x_1, x_2, .., x_d)}{\Delta x_2}$$

If these limits exist.

Schwarz/Clairaut theorem

Let $f\colon \mathcal{D} \to \mathbb{R}$ have continuous second order partial derivatives at a point $\boldsymbol{x}_0 = (x_1, x_2, \dots, x_d) \in \mathcal{D}$.
Then for all $1 \le i, j \le d$,

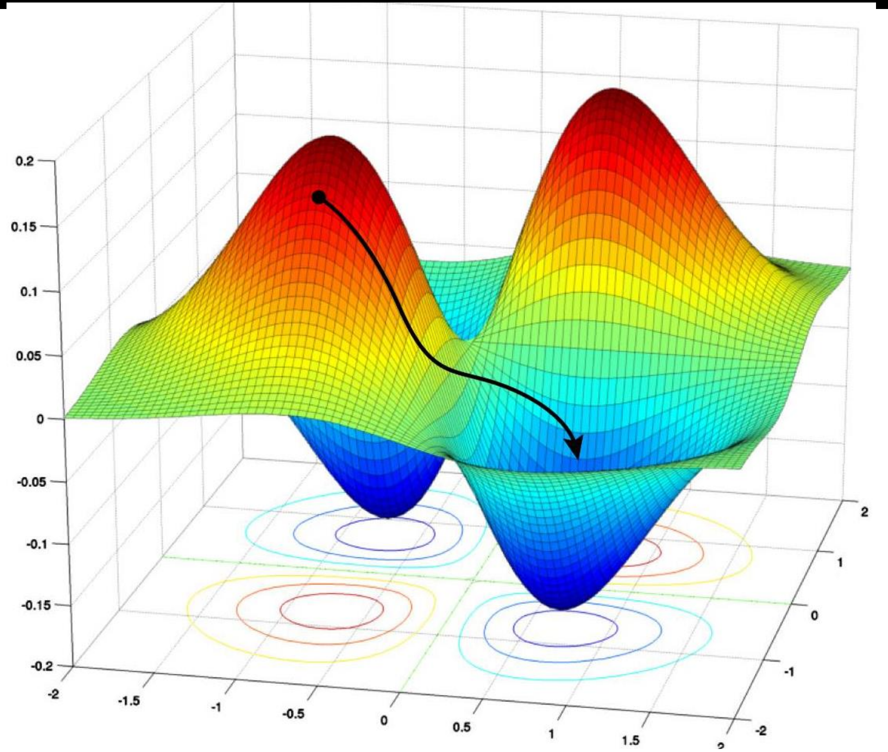
$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x_0}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\boldsymbol{x_0})$$

The chain rule

Let the functions $\boldsymbol{x}(t,\cdot) = (x_1(t,\cdot), .., x_d(t,\cdot))$ depend on t.

Consider a function $f: \mathcal{D} \to \mathbb{R}$.

How does $f(\boldsymbol{x}(t,\cdot))$ change with t?



source: sciencesprings.wordpress.com

Chain rule

If $f, \boldsymbol{x}$ are all continuously differentiable, then

$$\frac{\partial}{\partial t}\Big(f\big(\boldsymbol{x}(t,\cdot)\big)\Big) = \sum_{1 \le i \le d} \frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t}$$

The gradient

Gradient

If $f: \mathcal{D} \to \mathbb{R}$ is differentiable at a point $(x_1, .., x_d)$

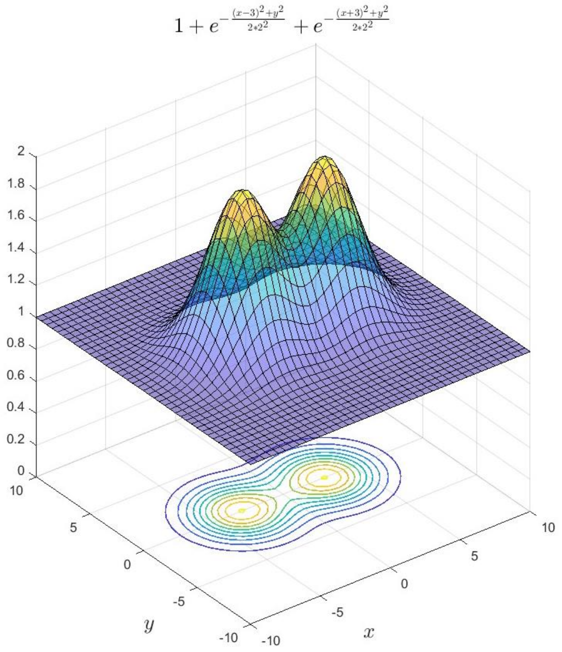
$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d}\right)$$
$$= \sum_{1 \le i \le d} \frac{\partial f}{\partial x_i} \, \hat{\boldsymbol{e}}_{x_i}$$

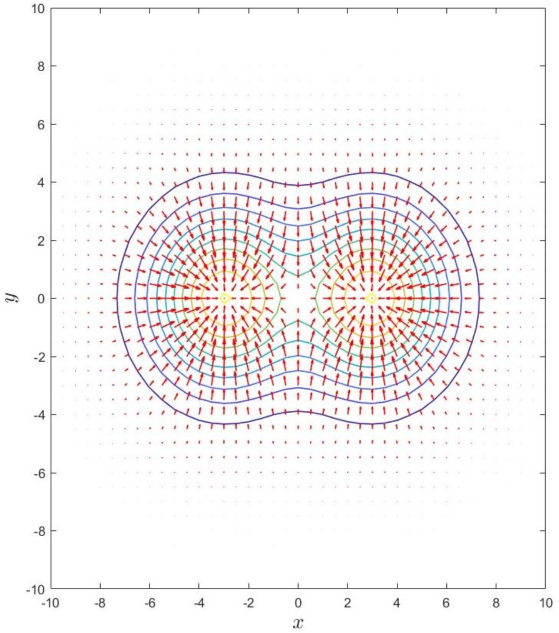
- ∇f is also commonly denoted by $\mathrm{grad}(f)$

Chain rule (revisited)

If $f, \boldsymbol{x}$ are all continuously differentiable, then

$$\frac{\partial}{\partial t}\Big(f\big(\boldsymbol{x}(t,\cdot)\big)\Big) = \nabla f\big(\boldsymbol{x}(t,\cdot)\big) \cdot \frac{\partial \boldsymbol{x}}{\partial t}$$





Run **grad_demo.m** to plot the function

$$1 + e^{\frac{(x-3)^2+y^2}{2*2^2}} + e^{\frac{(x+3)^2+y^2}{2*2^2}}$$

its contour and its gradient. What is the relation between the gradient and contour?

The gradient

Gradient

If $f: \mathcal{D} \to \mathbb{R}$ is differentiable at a point $(x_1, .., x_d)$

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d}\right)$$
$$= \sum_{1 \le i \le d} \frac{\partial f}{\partial x_i}\, \hat{\boldsymbol{e}}_{x_i}$$

- The gradient is perpendicular to the level sets
- The ∇f is along the direction in which f increases the fastest
- The maximum rate of increase is $|\nabla f|$

Questions 1.1

Let $f(x, y) = \frac{C}{\sqrt{x^2+y^2}}$.
Find ∇f and $|\nabla f|$.
Do you see a connection of this with physics?

Questions 1.2

Let $f(x, y) = \ln \sqrt{x^2 + y^2}$.
What is the tangent line of the level set of this curve at (x_0, y_0)?

Note

In cylindrical coordinate
$\nabla f = \frac{\partial f}{\partial r}\hat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{\partial f}{\partial z}\hat{\boldsymbol{k}}.$
What will be the formula in spherical coordinate?

Tangent planes revisited

- The tangent plane of a surface $z = f(x, y)$ at a point (x_0, y_0) is

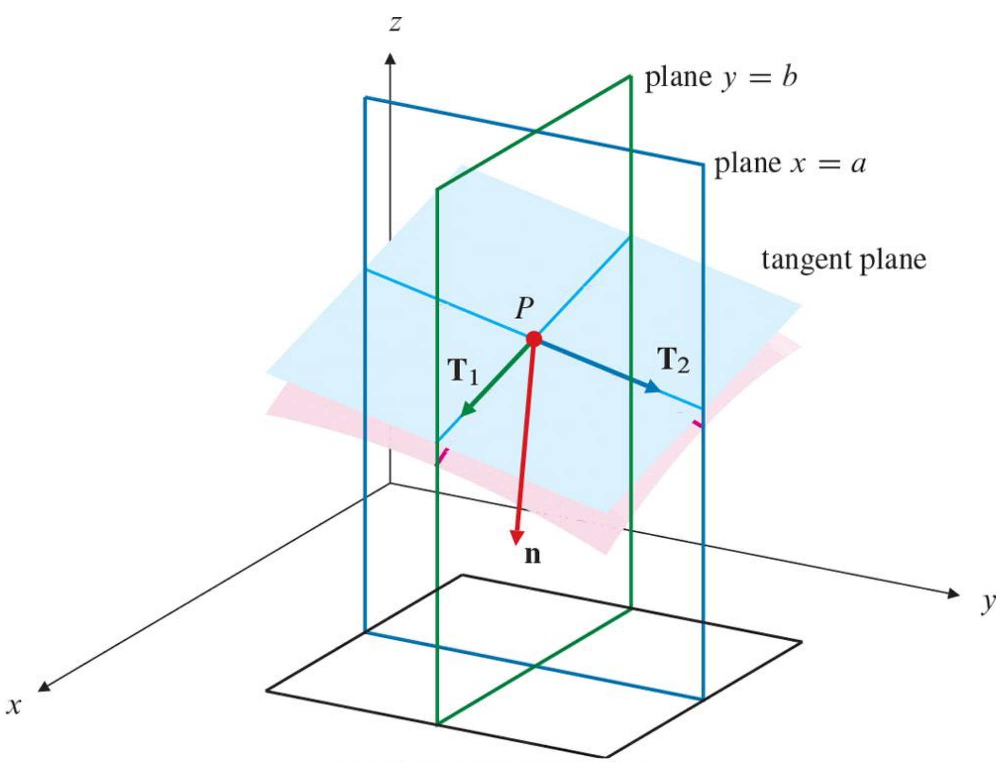
$$\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) - \left(z - f(x_0, y_0)\right) = 0$$

Or

$$z - f(x_0, y_0) = \nabla f(x_0, y_0) \cdot (\boldsymbol{x} - \boldsymbol{x}_0)$$

- The normal to this plain at point (x_0, y_0) is

$$\left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0), -1\right) = (\nabla f(x_0, y_0), -1)$$



Question 2

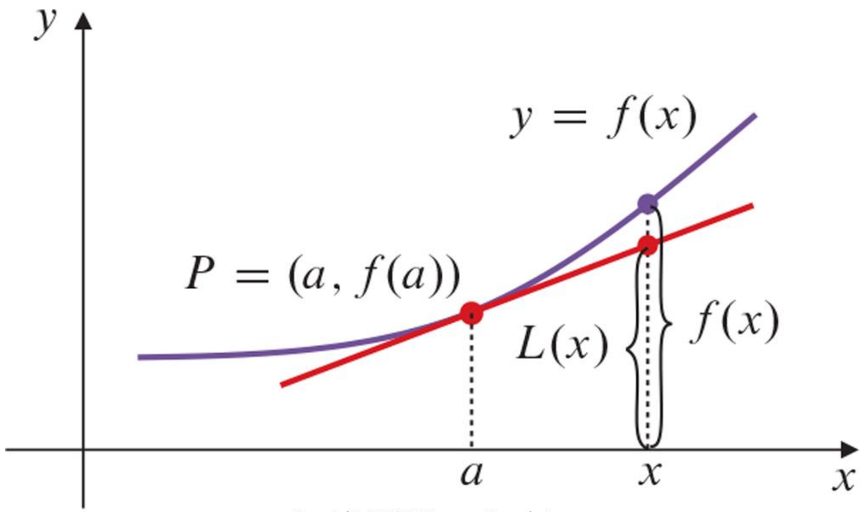
What is the tangent line at $(1,1,\sqrt{2})$ on the intersection curve of the cylinder $x^2 + (y-1)^2 = 1$ and the sphere $x^2 + y^2 + z^2 = 4$?

Linear approximation

Mean value theorem

Let $f: \mathcal{D} \to \mathbb{R}$ have continuous partial derivatives, $\boldsymbol{x}_0, \boldsymbol{x}_1 \in \mathcal{D}$ such that the points connecting them lies in $\mathcal{D}$.Then, there exists $\theta \in [0,1]$,

$$f(\boldsymbol{x}_1) - f(\boldsymbol{x}_0) = \nabla f\big(\boldsymbol{x}_0 + \theta(\boldsymbol{x}_1 - \boldsymbol{x}_0)\big) \cdot (\boldsymbol{x}_1 - \boldsymbol{x}_0)$$



- Another version of the theorem that does not require the line connecting $\boldsymbol{x}_0, \boldsymbol{x}_1$ to lie on $\mathcal{D}$ is given in Theorem 3.

Linear approximation

Let $|\boldsymbol{x}_1 - \boldsymbol{x}_0| \ll 1$. Then

$$f(\boldsymbol{x}_1) - f(\boldsymbol{x}_0) \approx \nabla f(\boldsymbol{x}_0) \cdot (\boldsymbol{x}_1 - \boldsymbol{x}_0)$$

Questions 3

What is the approximate value of the function

$$f(x, y) = \sin(\pi x y + \ln y)$$

at $(x, y) = (.05, 1.1)$?

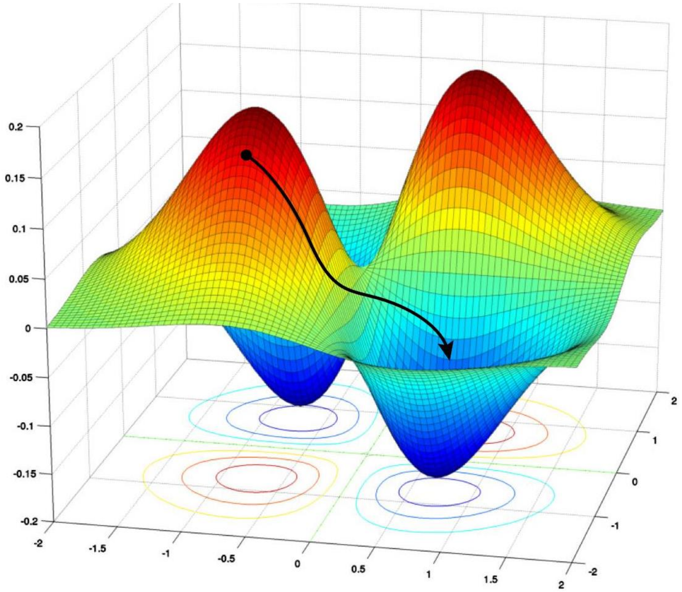
Directional derivatives

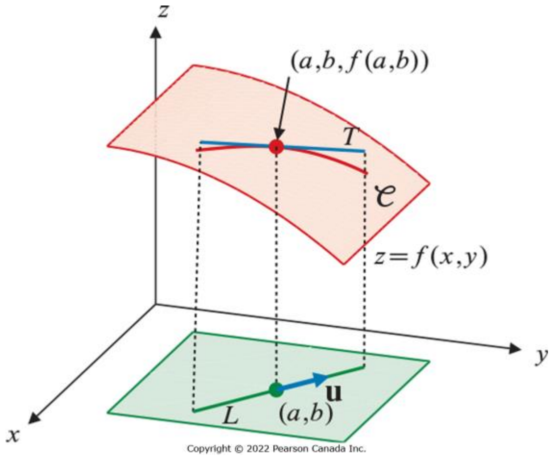
Directional derivatives

Directional derivative along a unit vector $\hat{\boldsymbol{u}} = (u_1, .., u_d)$.

$$D_{\hat{\boldsymbol{u}}} f(\boldsymbol{x}) = \lim_{t \to 0} \frac{f(x_1 + tu_1, x_2 + tu_2, .., x_d + tu_d) - f(x_1, x_2, .., x_d)}{t}$$

$$= \lim_{t \to 0} \frac{f(\boldsymbol{x} + t\,\hat{\boldsymbol{u}}) - f(\boldsymbol{x})}{t} = \nabla f(\boldsymbol{x}) \cdot \hat{\boldsymbol{u}}$$





Questions 4.1

Prove that along ∇f indeed f increases at the fastest rate with magnitude $|\nabla f|$.

Questions 4.2

Recall the mountain profile

$$H(x, y) = \sin(x)\sin(y).$$

You are hiking along the path $y = x^2 - 2$. What is the rate of elevation change per horizontal distance at (-1,-1)?

Vector fields

- Observe that the function $\nabla f(\boldsymbol{x})$ points $\boldsymbol{x} \in \mathbb{R}^d$ to $\nabla f(\boldsymbol{x}) \in \mathbb{R}^d$.

This motivates us to introduce the concept of vector fields

Vector fields

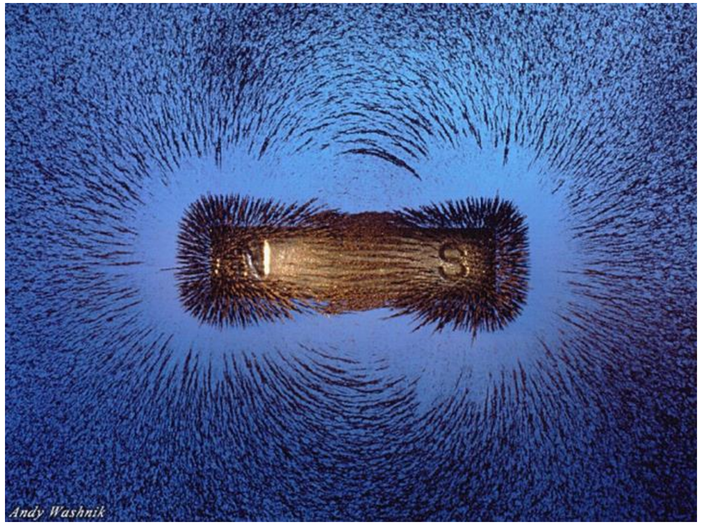
For a domain $\mathcal{D} \subseteq \mathbb{R}^d$, a vector field $\boldsymbol{f}$ is a function of the form

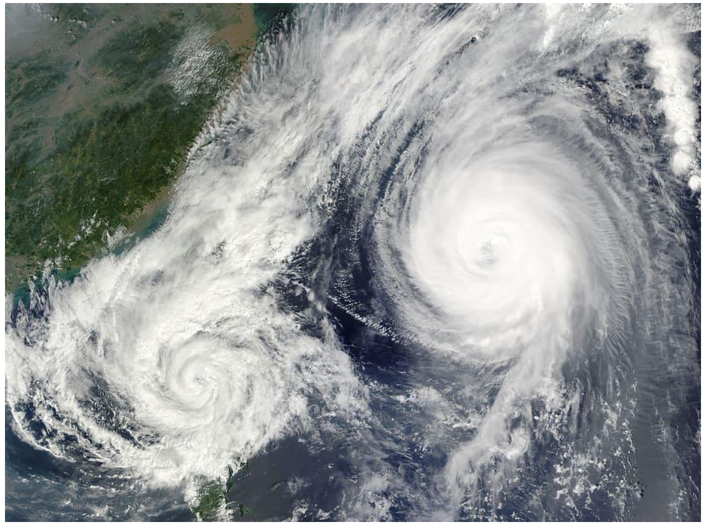
$$\boldsymbol{f}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \dots, f_n(\boldsymbol{x}))$$

written also as

$$\boldsymbol{f}: \mathcal{D} \to \mathbb{R}^n$$

Run **vector_field.m** and make a quiver-plot of a given vector field





The Jacobian

- What is the differentiation of a vector field?

The Jacobian

For a domain $\mathcal{D} \subseteq \mathbb{R}^d$, consider the vector field $\boldsymbol{f}: \mathcal{D} \to \mathbb{R}^n$ and let it be continuous at $x \in \mathcal{D}$. Then the Jacobian is the $n \times d$ matrix

$$D\boldsymbol{f}(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\boldsymbol{x}) & \dots & \frac{\partial f_1}{\partial x_d}(\boldsymbol{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(\boldsymbol{x}) & \dots & \frac{\partial f_n}{\partial x_d}(\boldsymbol{x}) \end{pmatrix}$$ provided all the derivatives exist.

- $D\boldsymbol{f}$ is also denoted by $\frac{\partial(f_1,\dots,f_n)}{\partial(x_1,\dots,x_d)}$ and $\nabla \boldsymbol{f}$

Chain Rule

Let $\boldsymbol{f}: \mathbb{R}^d \to \mathbb{R}^n$ and $\boldsymbol{g}: \mathbb{R}^n \to \mathbb{R}^m$ be differentiable functions. Then,

$$D(\boldsymbol{g} \circ \boldsymbol{f})(\boldsymbol{x}) = D\boldsymbol{g}\big(\boldsymbol{f}(\boldsymbol{x})\big) D\boldsymbol{f}(\boldsymbol{x})$$

Questions 5

Find the Jacobian for the following functions

- ❑ Cylindrical → Cartesian transform
  $(x, y, z) = (r\cos\theta, r\sin\theta, z)$
- ❑ Spherical → Cartesian transform
  $(x, y, z) = (R\sin\phi\cos\theta, R\sin\phi\sin\theta, R\cos\phi)$
- ❑ $\boldsymbol{g}(x, y, z) = \left(\ln(x^2 + yz), \sin\frac{xy}{z}, e^{-(x^2+y^2+z^2)}\right)$